Lihi Ancikovsky

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Education

Virginia Tech Blacksburg, VA

B.S. in Computer Science, Specialization in Data-Centric Computing, Mathematics Minor

- **GPA: 3.74 in Major** | Dean's List: Spring 2024, Spring 2025, Fall 2025, Spring 2026

Experience

Software Engineering Intern Jun. 2026 – Aug. 2026

Naval Surface Warfare Center Dahlgren Division

- Owned end-to-end development of a full-stack feature, turning a hidden, multi-step manual process into a one-click, near-instant workflow by designing the Angular/TypeScript frontend and implementing backend controller, business, and repository layers in a distributed, RabbitMQ-based microservices architecture.
- Engineered Angular/TypeScript interfaces backed by SQL and Elasticsearch, with RxJS-driven asynchronous data pipelines, lazy loading, and secure routing to simplify mission planning workflows.
- Proactively expanded the feature beyond its original scope, meeting frequently with the product owner to refine requirements and presenting progress to the full department and engineering leadership.
- Delivered production-quality software with 100% unit test coverage and rigorous code reviews, deploying within the team's Kubernetes/Docker infrastructure.

Artificial Intelligence Researcher Sep. 2025 – Present

Hume Center for National Security and Technology, Virginia Tech

- Build and train AI models for missile defense applications, design evaluation metrics, and develop reproducible experiments; use JaxMARL to train multi-agent reinforcement learning models across distributed defense scenarios.
- Generate leaderboards, plots, and summary tables for sponsor reporting; develop interpretable metrics evaluating model accuracy, resource efficiency, and performance stability across varying battery configurations.
- Presented a research poster, "Multi-Agent Reinforcement Learning for Distributed Missile Defense," at the Hume Center Colloquium (April 2026) to faculty, sponsors, and outside company and research representatives.

Data Science Intern, Manufacturing Dept. Jul. 2025 – Aug. 2025

Neocis

- Led a company-requested investigation into low component-pairing yield for a robotic dental surgery system, where mass-produced drill head/body components (not designed for robotic-arm precision) needed tight tolerance matching, and only 15% were pairing successfully.
- Partnered with manufacturing, quality assurance, software, and systems engineering teams to analyze the pairing algorithm using statistical testing in Python and JMP, tracing the shortfall to a misinterpreted threshold parameter in the pairing logic.
- Presented root-cause findings to engineering leadership in a stakeholder review, clarifying a misunderstood parameter and securing approval for a corrected value that increased valid component pairing from 15% to 50% and reduced inventory waste.

Extracurriculars and Awards

HackViolet, Virginia Tech Feb. 2025 – Present

Executive Director (May 2026 – Present), Judging Chair (May 2025 – May 2026), Peer Mentor and Winner (Feb. 2025)

- Set strategic direction and lead hackathon operations on a $15K+ budget, projecting 30% higher sponsorship revenue and growth to 500+ participants.
- Build and lead a high-performing executive board while cultivating sponsor, university, and partner-organization relationships to secure funding and event support.
- Oversee hackathon judging and training; overhauled the scoring platform and introduced new prize categories for efficiency and fairness.

Girls Who Code, Virginia Tech Aug. 2023 – Present

President (May 2026 – Present), Vice President (Jan. – May 2026), Treasurer (Jan. – Dec. 2025), Mentor (Aug. 2023 – Jan. 2025)

- Maintain the club website and email list, source and share opportunities, and run weekly meetings and workshops for 30+ regular members.
- Previously managed the club budget and sponsorships as Treasurer and mentored peers on coding and career development as a Mentor, supporting women in computing.

Projects

Cookie Crumblr | Hackathon Winner (2nd Overall) Mar. 2026

Python, JavaScript, React, AWS, Firefox WebExtensions API, Machine Learning

- Built a Firefox browser extension that protects users from ad-tech tracking by classifying cookies in real time with a trained K-Nearest Neighbors model (k=11, std_dev 0.05866) and automatically scrambling harmful tracking cookies while preserving their size and encoding.
- Engineered a full-stack pipeline connecting the browser extension to an AWS-hosted backend for real-time cookie classification, plus a React dashboard displaying cookie status, safety metrics, and user controls.

SpeakEasy: AI-Powered Personal Coach | Hackathon Winner Sep. 2025

Python, PyTorch, Flask, React, OpenCV, librosa, NumPy, MoviePy, Whisper, Gemini API

- Built a full-stack multi-modal AI system that analyzes public speaking videos and gives real-time feedback on pacing, clarity, gestures, tone, and delivery, using a deep learning pipeline fusing audio (prosody, MFCCs, filler detection) and video (facial landmarks, gesture tracking) signals.
- Developed and deployed a REST API with a React front end for video upload, live recording, and an interactive results dashboard, using the Gemini and YouTube APIs for reference retrieval and transcript analysis; placed among 115+ teams.

Technical Skills

Languages: Java, Python, C#, SQL, C, C++, JavaScript, HTML/CSS, MATLAB, x86, RISC-V, Rust, TypeScript, React
Frameworks/Tools: Angular, RxJS, Flask, Git, GitHub, MongoDB, Docker, Kubernetes, Elasticsearch, RabbitMQ, JMP, Jira
Libraries: NumPy, pandas, NetworkX, PyTorch, OpenCV, dlib, librosa, MoviePy, scikit-learn, Matplotlib
Skills: Deep Learning, Reinforcement Learning, Multi-Agent Systems, Multimodal Learning, NLP, Computer Vision, Speech Processing, Data Preprocessing, Model Training & Evaluation, REST APIs, Agile Development, Statistics, Data Structures and Algorithms, Data Analytics, Systems